

Monte Carlo Replication for Infection Rates

Robin Thrush and Kaitlynn Walston

December 17th, 2025

Introduction:

Beggs, et al. constructed a stochastic Monte Carlo model to better understand the transmission of airborne infection in a hypothetical hospital waiting room space with a volume of 132m^3 . Their working variables were all randomly sampled from a normal distribution. The authors explained how for each variable, they pulled means and standard deviations from other papers that had the empirical data ready and available. For example, the pulmonary rate of susceptible individuals was modeled with a mean of $0.48\text{m}^3/\text{h}$ and a standard deviation of $0.2\text{m}^3/\text{h}$. Other variables were held constant for each simulation such as exposure time, the number of infected individuals, room size, and air change rate. The actual equation was calculating the probability that susceptible individuals would contract the airborne disease. They simulated results for tuberculosis, measles, and influenza.

The goal for this project is to be able to replicate the author's stochastic Monte Carlo model. The model is stochastic in the sense that key variables are treated as random and for each simulation they are randomly sampled from the normal distribution. The randomness embodies the natural variability of the factors that influence airborne disease transmission. We want to change some of their constant variables like number of infected individuals, room size, and exposure time, in order to see how much effect each variable has on the predicted probability of infection for TB, measles, and the flu. Also, the extension portion of this project includes seeing what the model would predict for a whole different airborne disease (SARS) and how it is different or the same from the other diseases in its predicted infectious rate.

Assumptions in the study:

The researchers in the study set specific assumptions in order to make sure they could easily understand their experiment. The assumptions were that: only one infectious individual is present in their waiting room, the susceptible individuals can get infected but can not infect others, outside air is constantly being ventilated into the room, and air particles are a mixture of infectious and normal air and are evenly distributed throughout the room.

We wanted to mostly follow these assumptions, except the number of infectious people in the room. Keeping the other assumptions helped contain the infectious rate, where if the susceptible individual could infect others, the predicted probability of infection would grow exponentially.

Equation Explanation:

The equation that Beggs et al. used was taken further from an original probability equation made by Wells in 1955. Wells had defined quantum as a unit of infectious dose. The inhalation of one quantum droplets in the air and stated it would infect $1 - (1/e)$ ($\sim 63.2\%$) of the exposed animals. So, if 1 quantum is inhaled, the probability of infection to the person is 63.2% . In 1997 the Wells equation was extended by two researchers to build the Gammaitoni-Nucci version. In their version they accounted for finite exposure time, ventilation rates, and quanta concentration in indoor spaces. Beggs et al. implemented the Gammaitoni-Nucci since they were testing the airborne diseases in a hypothetical indoor space. Beggs et al. drew quanta rates for the different diseases from studies that estimated the rates. These quanta rates were essential for the model

as played a main role in predicting the probability of infection in the Monte Carlo Simulations: TB(12.7), Measles(570), and Flu(100).

Components of the EQN:

P: probability of infection on susceptible individuals S: number of susceptible individuals p: pulmonary ventilation rate I: number of infected individuals V: room volume N: room ventilation rate t: exposure time to susceptible individuals n: number of quanta of infection in the air the quantum generation rate : the total rate of quanta generation by all infectors

Replicating the Model Process:

As previously stated, Beggs et al. wrote a detailed paper, making the process of replicating their simulation smooth. In trying to follow this, we started by defining: a function has parameters for the number of simulations, the mean quanta rate, the standard deviation of the quanta rate, room volume, the number of infected individuals, and the room ventilation rate. We needed variables that keep track of: the probability of infection for the susceptible individuals, the number of newly infected individuals, and the number of susceptible individuals for each rep of the simulation. Inside our for loop, we define the variables: pulmonary ventilation rate, the quantum generation rate, the exposure time, initial number of susceptible individuals at time = 0. The 4 variables mentioned above are set to be our normally distributed randomly generated values (using *rnorm()*), with each of their means and standard deviations defined from the study. We define total rate of quanta generation by all infectors (quanta/min) by multiplying out number of infected individuals by the quanta rate. We then constructed the equation and at the end of 10,000 simulations we had a vector of the probability of infection for susceptible individuals. We took the mean of the probability. This was able to get our estimates that were similar to Beggs. study. This can be seen in the table below.

##		TB TB_vals	Flu Flu_vals	Measles Meas_vals
## 1	Beggs_TB	0.00340	Beggs_Flu 0.0262	Beggs_Meas 0.1349
## 2	Our_TB	0.00328	Our_Flu 0.0259	Our_Meas 0.1340

SARS Findings:

On top of trying to replicate their monte carlo model, we also wanted to test out the model with another infectious disease. We researched the SARS disease. *Mikszewski et al* wrote in a National Library of Medicine article that the estimate of infectious rate is about 2.7 quanta per hour with a standard deviation of 1.2 quanta. We could see in our results that the predicted mean probability resulted in a probability of 0.153, and had a moderate variability range of 0.072 to 0.369.

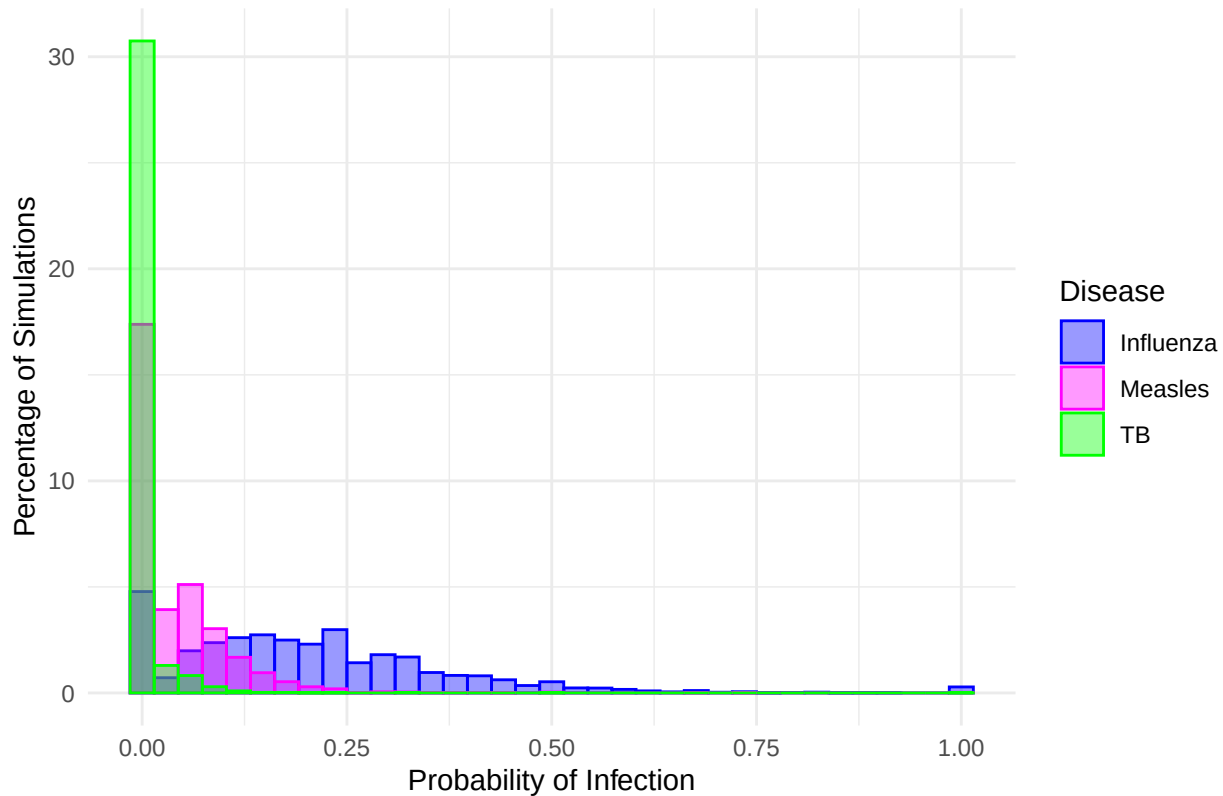
Scenario 1 & 2 Findings:

We wanted to create 2 different scenarios where we can change a few variables in order to see what variables were the most impactful in effecting the infectious prediction rate. For the first scenario, we wanted to increase the room volume, as well as the exposure time. We did this to test if increasing the room size would decrease the infectious rate, or if the longer exposure time, regardless of the room size, would have a larger effect. We increased the room volume size from 132 m^3 to 300 m^3 . We also increased the exposure rate of susceptible individuals by the infected from 30 seconds to 60 minutes. As a result of this experiment, our results showed that probability infection rate stayed very low, where most of our simulations stayed around 0-5% and barely other simulations reached past 25%. This showed us that the room volume was more impactful to the infection rate over the exposure time, since the infection rate did not increase.

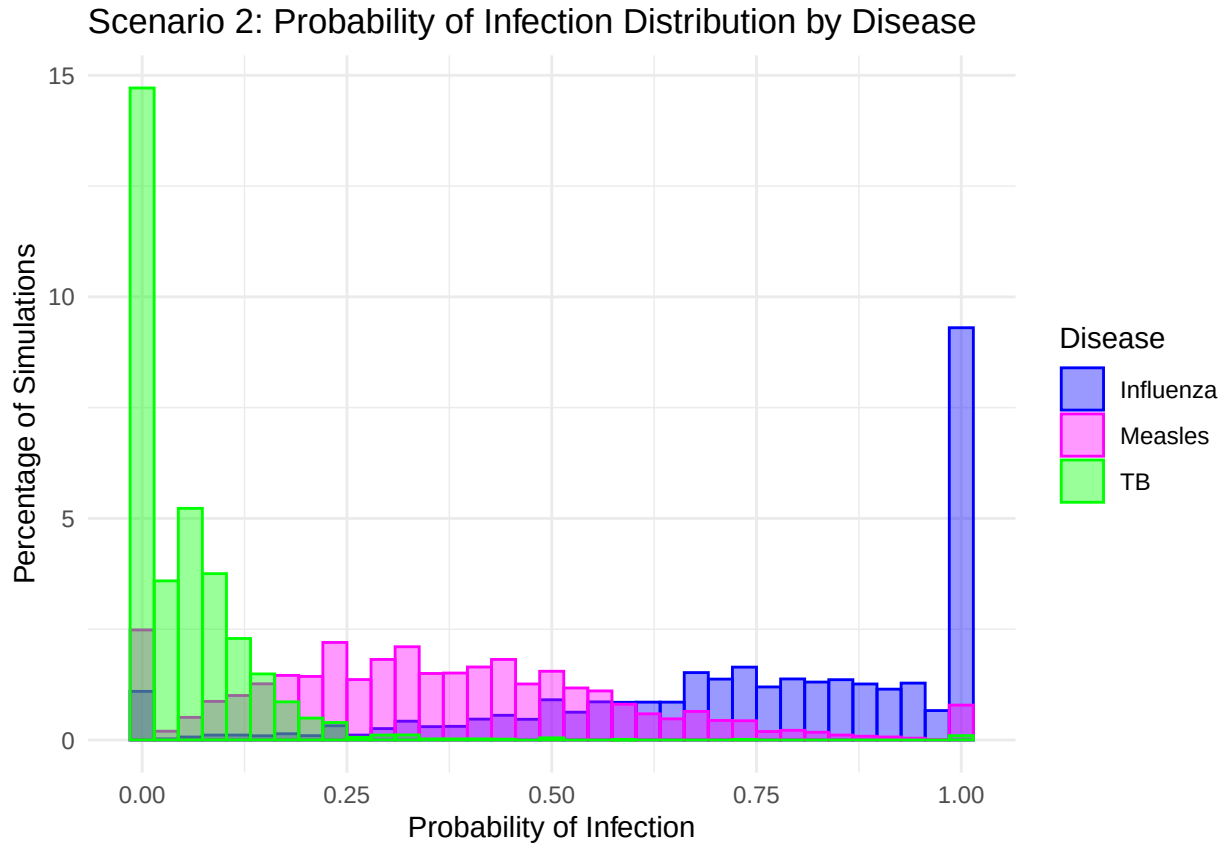
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our results showed that there was very small probability in each of the airborne diseases' simulations. The highest we saw probability wise was influenza at around 50% infection rate. Tuberculosis and Measles kept an infection rate at around a 0 to 25% infection rate. This shows us that the increased room size had more of an effect on the predicted probability than the exposure time, since being exposed longer did not lead to more infection.

Scenario 1: Probability of Infection Distribution by Disease



For the second scenario, we wanted to keep the original tested room volume size of 132 m^3 , as well as the original exposure time at 30 minutes. However we wanted to increase the number of infected individuals from one person to two people. We wanted to do this to see how much increasing the number of infected people would increase the infection rate. We can see from the results that there was in fact a higher probability of infection with influenza and measles tests. Tuberculosis stayed at a fairly small infection rate, but still saw a bit higher in some of the probability simulation results. This shows us that the number of infected people in the room greatly effected the prediction of infection, since the amount of volume and exposure time stayed the same as what the researchers did in the study.



Conclusion:

After completing our own replication of the researchers' Monte Carlo model, our take away was that the higher the quanta rate that an airborne disease has, or the higher the infection rate, the more impact it has on infecting people, regardless of other factors like room size, exposure time, and infected personnel. One limitation that we had when faced with everything was that neither of us have a lot of scientific knowledge and understanding. A lot of background research had to go into what even the researchers were talking about, to be able to break down into ideas we could understand and try to replicate. It was also difficult to imaging the room volumes and being able to tell of the room size they used was even a normal sized room or not. Overall, most of our predictions were correct where the number of infected people has more of an infection rate over changing the room size or exposure time.